

Supplementary Appendix for:
If a Statistical Model Predicts That Common
Events Should Occur Only Once in 10,000
Elections, Maybe it's the Wrong Model*

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S1 Statistical details

We provide technical details for Section 2. In particular, we provide the densities for the Normal, Student- t , and additive-logistic- t (S1.1), derive the probability for a Democratic victory as implied by our model (S1.2), detail the likelihood contributions of an election in district i and year t under the different cases of uncontestedness (S1.3), explicitly write out the posterior and each of our parameters (S1.4), describe our time indexing (S1.5), the exact covariate encoding (S1.6), and the sampler (S1.7). We open by collecting the paper’s reference tables: the time indices (Table S1), the location covariates (Table S2), the priors (Table S3), and the variable encodings (Table S4).

Table S1: Time indices. The lag $b(t)$ is the previous House election in both chambers: a Senate race’s lagged vote is its state’s House vote two years earlier, and the seat’s own previous race, six years earlier, enters as L^{sen} . The redistricting mapping $d(t)$ is characterized by $d(t) = d(t')$ if and only if year t and year t' operate under the same redistricting map.

Index	Meaning	Value
$m(t)$	midterm indicator	1 if t is a midterm, else 0
$d(t)$	redistricting decade of t	1940s–2020s
$b(t)$	the previous House election	$t - 2$
$p(t)$	most recent prior presidential election	$t - 4$ (presidential year), $t - 2$ (midterm)

S1.1 Densities

A normal distribution with mean m and scale s has support $(-\infty, \infty)$ and has density

$$f_N(x \mid m, s) = \frac{1}{s\sqrt{2\pi}} \exp\left\{-\frac{(x - m)^2}{2s^2}\right\}, \quad x \in \mathbb{R}. \quad (\text{S1})$$

The Student- t distribution with location m , scale $\phi > 0$, and degrees of freedom $\nu > 0$ has support $(-\infty, \infty)$ and density

$$f_t(x \mid m, \phi, \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}\phi} \left(1 + \frac{1}{\nu}\left(\frac{x - m}{\phi}\right)^2\right)^{-\frac{\nu+1}{2}}, \quad x \in \mathbb{R}, \quad (\text{S2})$$

where $\Gamma(\cdot)$ denotes the gamma function, given by the integral

$$\Gamma(a) = \int_0^\infty t^{a-1} e^{-t} dt$$

Table S2: The location covariates. Each enters μ_{it} in equation (3). “Slope” is whether the coefficient is a fixed grand mean or a grand mean plus a random-walk deviation. The lagged vote together with the lagged presidential, Senate, and gubernatorial measures are the measures of standing partisanship. (Priors are in Table S3. Exact coding is in Table S4.) A check marks a covariate added since the standard model of Gelman and King (1994). The scale and degrees of freedom carry further covariates (open seat, polarization, single-district/Senate) with no counterpart in that model’s single error variance (Section 2.2.3).

Symbol	Encodes	Slope	New
$y_{i,b(t)}^*$	lagged (logit) two-party vote	year-varying	
I	incumbent on the ballot, signed by party	year-varying	
S	Southern state	year-varying	
L^{pres}	lagged district presidential lean	year-varying	✓
U	uncontested (current race)	year-varying	
O^{inv}	involuntary open seat	fixed	✓
V	voluntary open seat, signed by departing party	fixed	✓
A_d	incumbent-party premium, by decade	fixed	✓
E_d	exit-party premium, by decade	fixed	✓
L^{sen}	lagged Senate vote in the district (down-ballot)	fixed	✓
L^{gov}	lagged gubernatorial vote in the district (down-ballot)	fixed	✓
M	midterm penalty (president’s party)	fixed	

for $a > 0$. Throughout the paper, we write $t_\nu(z) = f_t(z \mid 0, 1, \nu)$ and let F_{t_ν} denote the respective cumulative distribution function.

The inverse logit transform of the Student- t yields the additive logistic- t (ALT) distribution. Formally, if $Y \sim t(\mu, \phi, \nu)$ then $V = \text{logit}^{-1}(Y)$ is distributed ALT, with support $(0, 1)$. We apply the change of variables to $y = \text{logit}(v) = \log\{v/(1 - v)\}$, giving a Jacobian of $dy/dv = 1/\{v(1 - v)\}$, yielding the density for V as

$$\text{ALT}(v; \mu, \phi, \nu) = f_t(\text{logit}(v) \mid \mu, \phi, \nu) \cdot \frac{1}{v(1 - v)} = \frac{1}{\phi v(1 - v)} t_\nu\left(\frac{\text{logit}(v) - \mu}{\phi}\right), \quad (\text{S3})$$

Taking $\nu \rightarrow \infty$, the limiting case of the ALT is the logit-normal. Because the inverse logit is a one-to-one transform, Section 2 writes the outcome as (2); on the logit scale this is $y_{it}^* \sim t_{\nu_{it}}(\mu_{it}, \phi_{it})$, which we use throughout the paper.

Table S3: Priors for the LogisTiCC model. $t(m, s, \nu)$ is a Student- t with location m , scale s , and ν degrees of freedom. Half- t is its zero-truncated form. $N(m, s^2)$ is a normal with mean m and variance s^2 . $\text{Exp}(r)$ is an exponential with rate r . Decade fixed effects are indexed over redistricting decades. Supplementary Appendix S7 shows that results are not driven by these priors.

Parameter	Role	Prior
Location μ_{it} (grand mean of each slope)		
β_{y^*}	lagged (logit) two-party vote	$t(0.13, 0.25, 5)$
β_I	incumbent on the ballot	$t(0, 0.5, 5)$
β_S, β_U	Southern state; uncontested	$t(0, 0.5, 5)$
$\beta_{L^{\text{pres}}}$	lagged district presidential lean	$t(0, 0.3, 5)$
$\beta_{O^{\text{inv}}}$	involuntary open seat	$t(-0.01, 0.07, 5)$
β_V	voluntary open seat	$t(0, 0.5, 5)$
β_{A_d}, β_{E_d}	party premiums, by decade	$t(0, 0.5, 5)$
$\beta_{L^{\text{sen}}}, \beta_{L^{\text{gov}}}$	lagged Senate and gubernatorial votes	$t(0, 0.2, 5)$
β_M	midterm penalty	$N(-0.3, 0.025^2)$
Random effects		
σ_γ	candidate-effect scale	Half- $t(0, 0.08, 40)$
$\sigma_{\eta,d}$	national-trend amplitude, by decade	Half- $t(0, 0.12, 5)$
σ_{β_k}	per-step innovation SD (year-varying slopes)	Half- $t(0, 0.03, 5)$
Scale ϕ_{it}		
ζ	intercept	$t(0, 0.05, 5)$
ζ_d	decade fixed effects	$t(0, 0.15, 5)$
δ	open seat	$N(0.15, 0.07^2)$
	marginal distance from 50/50	$N(0, 0.3^2)$
	single-district/Senate	$N(0, 0.3^2)$
Degrees of freedom $\nu_{it} = 1 + \exp(\cdot)$		
α	intercept	$N(0, 0.5^2)$
α_d	decade fixed effects	$t(0, 0.15, 5)$
λ	open seat	$N(-0.15, 0.1^2)$
	marginal distance from 50/50	$N(0, 0.5^2)$
	single-district/Senate	$N(0, 0.3^2)$
Imputation		
σ_{imp}	lagged-vote imputation noise, predecessor race	$\text{Exp}(1/0.05)$
κ	regression imputation: intercept	$t(0.1, 0.05, 5)$
	lagged presidential lean	$t(0.4, 0.15, 5)$
	incumbency code of the previous race	$t(0.3, 0.08, 5)$
	Democratic party defending the seat	$t(0, 0.15, 5)$
σ_{reg}	lagged-vote imputation noise, regression	$\text{Exp}(1/0.12)$

Table S4: Variable encodings, grouped by the outcome parameter each variable enters: the location μ_{it} in the top block, the scale and degrees of freedom ϕ_{it} and ν_{it} in the bottom block.

Symbol	Code name	Coding
<i>Location covariates (enter μ_{it})</i>		
$y_{i,b(t)}^*$	lag.house	lagged effective two-party Democratic vote, logit scale: the district's House vote two years earlier, or for a Senate race the state's. Observed where available, else latent and imputed (S8)–(S9)
I	i2	+1 if a Democratic incumbent is on the ballot, −1 if a Republican, else 0 (two paired incumbents also code 0)
O^{inv}	openi	+1 Democratic / −1 Republican involuntary exit, else 0
V	openv_s	+1 if open by a Democrat's voluntary exit, −1 by a Republican's, else 0
A_d	incany_<decade>	1 on incumbent-held seats in redistricting decade d , else 0
E_d	openvany_<decade>	1 on voluntarily vacated seats in decade d , else 0
U	unc	+1 if only a Democrat runs, −1 if only a Republican, 0 if contested
S	south	1 if the district is in a former Confederate state, else 0
L^{pres}	dpres.lag	lagged district presidential lean at $p(t)$, logit scale
L^{sen}	lag.sen	most recent Senate two-party Democratic vote, logit scale; for a Senate race, its seat's previous race
L^{gov}	lag.gov	most recent gubernatorial two-party Democratic vote, logit
M	penalty	+1 midterm under a Democratic president, −1 Republican, 0 on-year
<i>Dispersion covariates (enter the scale ϕ_{it} and degrees of freedom ν_{it})</i>		
O	is_open_seat	1 if no incumbent runs (any open seat), else 0
W	dist_dpres.lag	$ \text{logit}^{-1}(L^{\text{pres}}) - \frac{1}{2} $, marginal distance from 50/50
G	cand_effect_split	1 for single-district states and the Senate, else 0

S1.2 Probability of Victory

Proposition 1 provides us with a compact method of deriving the probability of a Democratic victory (i.e., $v_{it} > 0.5$) using the ratio μ/ϕ , which underlies the win probabilities used throughout the paper; we provide the proof below.

Proposition 1. *Let $V = \text{logit}^{-1}(Y)$ where $Y \sim t(\mu, \phi, \nu)$, so $V \sim \text{ALT}(\cdot; \mu, \phi, \nu)$. Then*

$$\int_{1/2}^1 \text{ALT}(v; \mu, \phi, \nu) dv = \Pr(V > \tfrac{1}{2}) = F_{t_\nu}(\mu/\phi), \quad \int_0^{1/2} \text{ALT}(v; \mu, \phi, \nu) dv = F_{t_\nu}(-\mu/\phi).$$

Proof. Since the logit is a strictly increasing bijection mapping $(0, 1)$ to $\mathbb{R}$, it follows that $V > \frac{1}{2} \iff Y > 0$ (the probability of the Democratic candidate winning the associated election). Hence,

$$\int_{1/2}^1 \text{ALT}(v; \mu, \phi, \nu) dv = \Pr(V > \tfrac{1}{2}) = \Pr(Y > 0) = F_{t_\nu}(\mu/\phi).$$

A similar argument shows that the probability of the Republican candidate winning the associated election is

$$\int_0^{1/2} \text{ALT}(v; \mu, \phi, \nu) dv = F_{t_\nu}(-\mu/\phi).$$

S1.3 Likelihood

Section 2.2.4 provides a compact formulation for the likelihood by splitting it into product of density contributions from contested races and integral contributions from uncontested races. In this section, we decompose the likelihood further by considering for each election, not only whether or not that election is contested, but also if its lagged race is contested. The relevancy of this consideration lies in the fact that the lagged Democratic vote share for district i in election year t acts as a covariate for constructing μ_{it} . Since the lagged election is also either contested or uncontested, we can bin all elections into four categories: (CC) both the present and lagged elections are contested, (UC) the present election is uncontested but the lagged election is contested, (CU) the present election is contested but the lagged election is uncontested, and (UU) both the present and lagged elections are uncontested. A lagged vote is also unobserved when the race has no predecessor in our data, as in a redrawn district, and we place such races in CU or UU as well.

For district i in election year t , let $U_{it} = 1$ if the district is uncontested and a Democrat runs, $U_{it} = 0$ if the district is contested, and $U_{it} = -1$ if the district is uncontested and a Republican runs. Also, let $y_{i,b(t)}^*$ denote the lagged effective vote. Since $y_{i,b(t)}^*$ is necessary

for computing μ_{it} , we make this dependence notationally explicit by writing $\mu_{it}(y^*)$ when the lagged vote is unobserved. We factor our likelihood as

$$\mathcal{L} = \prod_{CC} \mathcal{L}_{it}^{CC} \prod_{UC} \mathcal{L}_{it}^{UC} \prod_{CU} \mathcal{L}_{it}^{CU} \prod_{UU} \mathcal{L}_{it}^{UU}. \quad (\text{S4})$$

Recall from Section 2.2.4 the likelihood contribution of present elections and the imputation method we introduced for uncontested lagged elections. Combining the two, it follows that the likelihood contributions for the four cases (UU), (UC), (CU), (CC) are given by

$$\mathcal{L}_{it}^{CC} = \frac{1}{\phi_{it}} t_{\nu_{it}} \left(\frac{y_{it}^* - \mu_{it}}{\phi_{it}} \right), \quad \mathcal{L}_{it}^{UC} = C_{it}(\mu_{it}), \quad (\text{S5})$$

$$\mathcal{L}_{it}^{CU} = \int \frac{1}{\phi_{it}} t_{\nu_{it}} \left(\frac{y_{it}^* - \mu_{it}(y^*)}{\phi_{it}} \right) g(y^*) dy^*, \quad \mathcal{L}_{it}^{UU} = \int C_{it}(\mu_{it}(y^*)) g(y^*) dy^*, \quad (\text{S6})$$

where C_{it} is the probability that the unopposed party's effective vote lies in the censoring interval of Section 2.2.4, between 50 and 99 percent,

$$C_{it}(\mu) = F_{t_{\nu_{it}}} \left(\frac{U_{it} \mu}{\phi_{it}} \right) - F_{t_{\nu_{it}}} \left(\frac{U_{it} \mu - A}{\phi_{it}} \right), \quad A = \text{logit}(0.99) = \log 99. \quad (\text{S7})$$

Here $g(y^*)$ is the conditional density of the unobserved lagged effective vote $y^* = y_{i,b(t)}^*$ given the parameters. Its form depends on whether the race that produced the lag is in our data. Let $\mathcal{O}$ be the set of races (i, t) whose lagged vote is observed; $\mathcal{S}$ the races whose lagged vote is unobserved and whose predecessor race, the last regular race on the same ground two years earlier, is in our data, written $\pi(i, t)$; and $\mathcal{F}$ the races whose lagged vote is unobserved and whose predecessor is not. In the full-data fit, $\mathcal{O}$ holds 13,330 races, $\mathcal{S}$ holds 1,965, and $\mathcal{F}$ holds 3,590: 3,504 in redrawn districts, 76 in 1946, the first year of our data, and 10 Senate races whose statewide House lag is missing. For $(i, t) \in \mathcal{S}$, the model imputes the lag from the predecessor race,

$$y_{i,b(t)}^* \sim N(\tilde{\mu}_{\pi(i,t)}, \sigma_{\text{imp}}^2), \quad (\text{S8})$$

where $\tilde{\mu}_{\pi(i,t)}$ is the linear predictor the model builds for the predecessor race (using its covariates, candidate, and year effects, but excluding its own lagged-vote term to avoid recursion), and σ_{imp} is a shared scale parameter for these imputations. For $(i, t) \in \mathcal{F}$, the

model imputes the lag from a regression,

$$y_{i,b(t)}^* \sim N(w_{it}\kappa, \sigma_{\text{reg}}^2), \quad (\text{S9})$$

where w_{it} holds an intercept, the lagged presidential lean L^{pres} , the incumbency code of the seat's previous race (`i1`), and an indicator that the party defending the seat is Democratic (`inc.party`), all known before the election. The regression is estimated on the observed lags, which contribute $\prod_{(i,t) \in \mathcal{O}} f_N(y_{i,b(t)}^* \mid w_{it}\kappa, \sigma_{\text{reg}})$ to the posterior, so an observed lag enters the model twice: as a covariate in μ_{it} and as an outcome of (S9). The parameters σ_{imp} , κ , and σ_{reg} are estimated jointly with the rest of the model (priors in Table S3).

Both (S8) and (S9) are untruncated auxiliary Normal laws for a missing covariate. For a race in $\mathcal{S}$ with an uncontested predecessor, the model's own law for the earlier effective vote would be a Student- t truncated to the censoring interval of (S7); that law is computationally taxing to evaluate inside the sampler, so the fitted model uses (S8) in its place. The censoring of an earlier race therefore enters the posterior only through that race's own contribution, $\mathcal{L}^{\text{UC}}$ or $\mathcal{L}^{\text{UU}}$. The imputations propagate the uncertainty about each missing lag to μ_{it} through the integrals in (S6).

S1.4 Posterior

Let $\Theta = \{\Theta_\mu, \Theta_\phi, \Theta_\nu\}$ partition the model parameters into blocks, where g delineates grouping parameters (such as year or candidate) and p denotes parameters that are pooled for the entire dataset. These parameters contribute to the location, scale, and degrees of freedom parameters of the outcome respectively. Furthermore, let $y_{\mathcal{S}}^*$ and $y_{\mathcal{F}}^*$ denote the latent lagged effective votes of the races in $\mathcal{S}$ and $\mathcal{F}$ (Supplementary Appendix S1.3). Then by Bayes' rule, the model posterior is proportional to

$$\begin{aligned} & p(\Theta, y_{\mathcal{S}}^*, y_{\mathcal{F}}^* \mid \text{data}) \\ & \propto \left[\prod_{i,t} \mathcal{L}_{it}(\Theta, y_{\mathcal{S}}^*, y_{\mathcal{F}}^*) \right] \prod_{(i,t) \in \mathcal{O}} f_N(y_{i,b(t)}^* \mid w_{it}\kappa, \sigma_{\text{reg}}) \\ & \times \prod_{(i,t) \in \mathcal{S}} f_N(y_{i,b(t)}^* \mid \tilde{\mu}_{\pi(i,t)}, \sigma_{\text{imp}}) \prod_{(i,t) \in \mathcal{F}} f_N(y_{i,b(t)}^* \mid w_{it}\kappa, \sigma_{\text{reg}}) \\ & \times \prod_c f_N(\gamma_c \mid 0, \sigma_\gamma) \prod_t f_N(\eta_t \mid 0, \sigma_{\eta,d(t)}) \prod_k \prod_{t > 1946} f_N(\beta_{tk} \mid \beta_{t-2,k}, \sigma_{\beta_k}) \\ & \times p(\Theta_p) p(\Theta_\phi) p(\Theta_\nu), \end{aligned} \quad (\text{S10})$$

where $\mathcal{L}_{it}$ is $\mathcal{L}_{it}^{\text{CC}}$ or $\mathcal{L}_{it}^{\text{UC}}$ from Supplementary Appendix [S1.3](#), with μ_{it} evaluated at the latent lag when the lag is unobserved, so that integrating over $y_{\mathcal{S}}^*$ and $y_{\mathcal{F}}^*$ yields the CU and UU contributions ([S6](#)), and f_N is the normal density ([S1](#)). The first row of factors holds the two likelihoods, of the current outcomes and of the regression ([S9](#)) on the observed lags; the second holds the two imputation laws, ([S8](#)) and ([S9](#)); and the third holds the three hierarchies of ([3](#)): the candidate effects γ_c of every candidate c , the national effects η_t , and a random walk for each year-varying slope k (Table [S2](#)). Each walk starts at its grand mean, $\beta_{1946,k} = \beta_k$, so β_k is the slope’s 1946 level and the walk has no free initial deviation. The last row is the product of the priors in Table [S3](#) on the blocks below.

The blocks are

$$\Theta_{\mu} = \Theta_g \cup \Theta_p, \quad \Theta_{\phi} = \{\zeta, \zeta_d, \delta\}, \quad \Theta_{\nu} = \{\alpha, \alpha_d, \lambda\},$$

where

$$\Theta_g = \{ \{ \gamma_D \}, \{ \gamma_R \}, \{ \eta_t \}, \{ \beta_{tk} \} \}, \quad \Theta_p = \{ \{ \beta_k \}, \sigma_{\gamma}, \{ \sigma_{\beta_k} \}, \{ \sigma_{\eta,d} \}, \sigma_{\text{imp}}, \kappa, \sigma_{\text{reg}} \}.$$

S1.5 Time indices

Throughout the paper, we regularly incorporate mappings $m(t), d(t), b(t), p(t)$ that take time as an argument for notational compactness. We detail the exact meaning of each of these mappings in Table [S1](#).

The lag $b(t)$ is the previous House election, two years earlier, in both chambers. In the House, all seats are up for re-election every two years, so $y_{i,b(t)}^*$ is the district’s own previous vote. In the Senate, a third of the seats is up for re-election every two years, so a seat’s own previous race is six years old; a Senate race therefore takes its state’s two-party House vote two years earlier as $y_{i,b(t)}^*$, and its seat’s previous race enters separately as L^{sen} . The presidential lag $p(t)$ identifies the most recent presidential election, which is either two years ago during midterm elections or four years ago during presidential elections. Its purpose is to identify the proper lagged presidential vote to include as a covariate for district i in election year t . In similar spirit, $m(t)$ serves as an indicator function that notes if an election occurs during a midterm. Finally, $d(t)$ maps years to redistricting decades, which

is key to identifying long-term structural changes to the variance of the national trend and in the scale and degrees of freedom parameters for the Student- t parameterization.

S1.6 Variable encoding

In Table S4, we report how we encode all variables.

S1.7 Sampler and diagnostics

The model is estimated through Hamiltonian Monte Carlo on a single NVIDIA H200 GPU (Supplementary Appendix S4), using the no-U-turn sampler implemented in NumPyro. We re-parametrize all random variables in non-centered form whenever possible to avoid unwanted funnels during the sampling process. The production fit to the full data uses 20 independent chains, run with 8,000 sampling draws each (160,000 draws in total) after 4,000 warmup iterations per chain, at target acceptance rate of 0.90 and maximum tree depth 13, using a diagonal mass matrix and overdispersed per-chain starts (for the $\hat{R}$), under a fixed seed. Posterior estimation took approximately 11.0 hours. We save a trace containing all 160,000 posterior draws, and report the diagnostics for the model run in Table S5. The other fits use the same sampler settings with fewer draws: the leave-future-out holdout fits behind the out-of-sample results, 20 chains of 5,000 draws after 2,000 warmup iterations; the ablation and build-up fits (Supplementary Appendices S2 and S3), 4 chains of 1,000 draws after 2,000; the doubled-prior fit to the full data (Supplementary Appendix S7), 4 chains of 4,000 draws after 4,000; and the chamber comparison, 40 chains of 4,000 draws after 4,000.

S2 Ablation Studies

We make four modeling innovations to achieve generatively accurate model predictions, described in Section 2.2.1 of the paper: parameter drift with coefficient stability, positive cross-district covariances through the national trend, district-level political surprises represented by heavy tails, and the additive logistic Student t distribution with race-specific scale and degrees of freedom. In this section, we conduct “ablation studies,” where each

Table S5: MCMC convergence diagnostics. $\hat{R}$ and the effective sample sizes (ESS) are computed over the 313 structural parameters. The sampler-level diagnostics (divergences, BFMI, tree depth, and acceptance) cover all 160,000 transitions.

Diagnostic	Value	Target
Divergent transitions	0 (0%)	< 0.1%
Max $\hat{R}$	1.002	≤ 1.01
Parameters with $\hat{R} > 1.01$	0/313	0
Minimum bulk ESS	12,232	≥ 400
Minimum tail ESS	19,323	≥ 400
BFMI (per-chain range)	0.42–0.49	> 0.30
Mean acceptance probability	0.947	≈ 0.90
Tree depth (median / max)	10/11 (cap 13)	< cap
Tree-depth saturation (≥ 13)	0.0%	< 5%

model component is removed in turn, holding everything else fixed, to show how the model degrades. The conclusion of this section is that all four components are essential to achieve the performance we report.

Each ablation is the published model with exactly one component switched off: (1) *no coefficient drift*, pinning the year-varying covariate effects to constants; (2) *no national trend*, pinning σ_η to zero so districts are conditionally independent within a year; (3) *no heavy tails*, pinning the degrees of freedom so the additive logistic Student t reduces to the additive logistic normal while keeping the modeled scale; and (4) *no dispersion structure*, pinning the scale and degrees-of-freedom regressions to constants so every race shares one ϕ and one ν . We re-fit each ablated model on the full data and again on the leave-future-out panels, and forecast the held-out elections exactly as for the full model.

Table S6 reports held-out district-level coverage, RMSE, and the continuous ranked probability score (CRPS; Gneiting and Raftery 2007), which scores each forecast’s whole predictive distribution against the observed vote (lower is better; Supplementary Appendix S10), and the count of one-in-10,000 events (observed votes outside the 99.99 percent interval, as in Figure 1 of the paper) for the full model, the four ablations, and the doubled-hyperprior sensitivity fit, pooled over every held-out election from 1950 through 2024 (14,478 House races; Senate races from 1960), and Figure S1 plots empirical against nominal coverage. Removing any component degrades held-out performance. Freezing

the coefficient drift raises held-out House RMSE by six percent and CRPS by six. Removing the dispersion regressions costs one percent on CRPS. Removing the national trend costs four percent on RMSE and five on CRPS pooled over all years, but the pooled numbers understate the component: the penalty is concentrated in exactly the elections the trend exists for (49 percent in 2010, 75 percent in 2004, and 59 percent in 2024, against about three percent in a typical year), and the cross-district correlation it induces is what widens legislature-level intervals to honest levels (Supplementary Appendix [S11](#)). Replacing the additive logistic Student t with the additive logistic normal barely moves RMSE (0.0725 against 0.0719) and raises CRPS by two percent, while pushing 95 percent coverage to 96.2 percent, because a normal likelihood cannot treat surprises as tail events and instead inflates every interval. Even with those wider intervals, the normal tails place 22 House races outside the 99.99 percent interval, against 2 for the full model and the 1.4 a calibrated model expects; the other three ablations place between 0 and 4. The doubled-hyperprior fit, by contrast, matches the published model’s accuracy (House RMSE 0.0719 against 0.0719; CRPS within one percent), which is the predictive form of the statement in Supplementary Appendix [S7](#), though its intervals are somewhat wider (97.2 percent coverage at the 95 percent level, against 95.8).

S3 Building the Model One Component at a Time

Each component of LogisTiCC improves held-out forecasts when we add the components one at a time, in the order of Section [2.2.1](#) of the paper. The ablations of Supplementary Appendix [S2](#) remove one component from the full model; Figure [S2](#) runs the exercise in reverse. It starts from the standard model of Section [2.1](#), the normal model of Table [S7](#), and adds (1) candidate effects and the logit transformation, with every election pooled, constant coefficients, no national trend, one error scale and normal tails; (2) parameter drift with coefficient stability; (3) the national trend; (4) district-level political surprises, which let the error scale vary with the characteristics of each race; and (5) the additive logistic Student t , whose degrees of freedom also vary by race and which completes LogisTiCC. We fit each intermediate model to every leave-future-out panel, forecast each

	50% cov.	80% cov.	95% cov.	RMSE	CRPS	1-in-10,000 events
<i>House</i> (n = 14,478 races, held-out 38 years; 1.4 1-in-10,000 events expected)						
Full model	51.3	81.4	95.8	0.0719	0.0381	2
No coefficient drift	48.3	78.7	95.1	0.0760	0.0405	1
No national trend	48.1	77.8	94.3	0.0748	0.0401	4
Normal tails	56.3	84.3	96.2	0.0725	0.0388	22
No dispersion model	53.3	82.2	96.1	0.0718	0.0385	0
Hyperprior scales $\times 2$	55.6	85.0	97.2	0.0719	0.0384	0
<i>Senate</i> (n = 1,104 races, held-out 33 years; 0.1 1-in-10,000 events expected)						
Full model	47.5	77.6	93.4	0.0891	0.0482	0
No coefficient drift	43.5	74.3	92.3	0.0940	0.0514	0
No national trend	45.4	75.1	92.8	0.0910	0.0496	0
Normal tails	53.6	80.7	93.4	0.0893	0.0490	4
No dispersion model	45.7	74.2	92.0	0.0887	0.0483	0
Hyperprior scales $\times 2$	50.1	81.2	95.3	0.0893	0.0485	0

Table S6: Ablation study: held-out one-step-ahead performance with each model component removed in turn, pooled over every held-out election from 1950 through 2024 (the Senate’s from 1960). The last column counts the observed votes outside the 99.99 percent interval, as in Figure 1 of the paper.

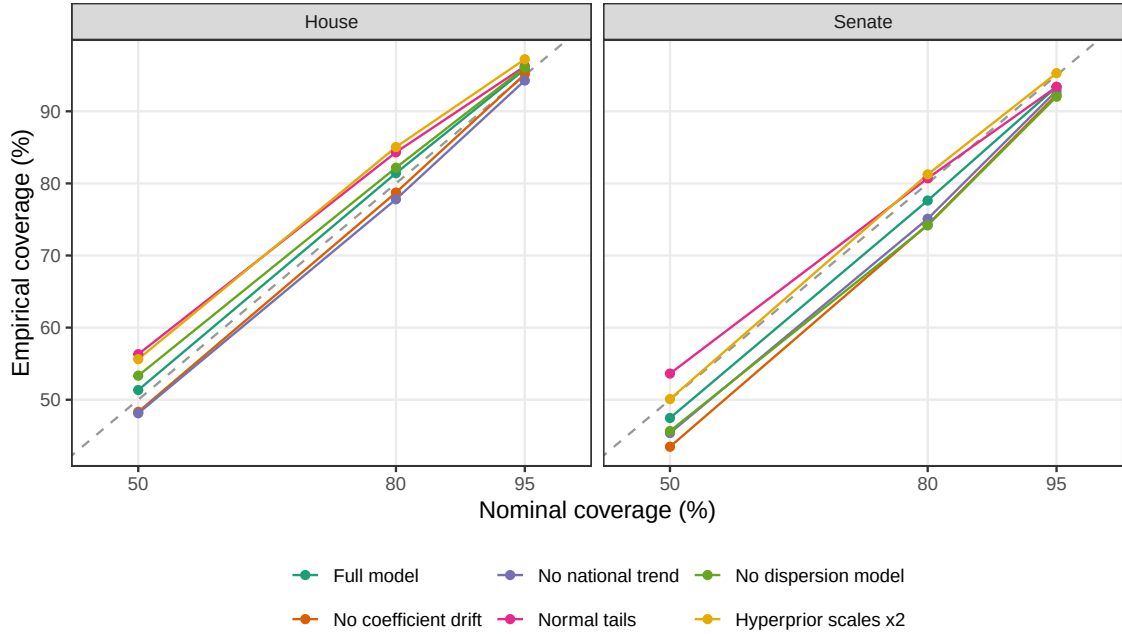


Figure S1: Empirical against nominal predictive-interval coverage, out of sample, for the full model and each ablation.

held-out election from 1950 through 2024 (the Senate’s from 1960) exactly as for the full model, and score every step’s CRPS on the same 11,436 contested House races and 1,104 contested Senate races, and its one-in-10,000 events on all the contested races it forecasts, as in Figure 1 of the paper (the standard model’s from the normal model of that figure).

Every component lowers the CRPS of House forecasts (panel (a) of Figure S2). Candidate effects and the logit transformation cut CRPS by 18.2 percent of the standard model’s, parameter drift by a further 1.9 points, the national trend by 0.5, district-level surprises by 1.4 and the additive logistic Student t by 2.2, for a total reduction of 24.2 percent. In the Senate, with a tenth as many races, the gains concentrate in the first two components (4.0 and 3.5 points of an 8.6 percent total), and the national trend and district-level surprises move CRPS by less than half a point.

Panel (b) turns to the tails of the forecast distribution, the focus of the paper. The standard model places 62 of its 11,436 House forecasts outside its 99.99 percent interval, where a calibrated model expects 1.1. Each component lowers that count, among the 14,478 House races the other steps forecast (1.4 expected): 55 with candidate effects and the logit transformation, 40 with parameter drift, 25 with the national trend, 22 with district-level surprises and 2 with the additive logistic Student t , the count of Figure 1 of the paper. The national trend shows why the two panels differ: it changes House CRPS by only 0.5 points but cuts the extreme misses from 40 to 25, as expected of a component that represents the swings shared by every district in an election. The additive logistic Student t is the only component that brings the count of one-in-10,000 events in line with calibration.

What each component adds depends on the order: a component added early can absorb some of the gains of the components after it. We follow the order of the paper, and the ablations of Supplementary Appendix S2 give the complementary measure, the loss from removing each component with all the others in place.

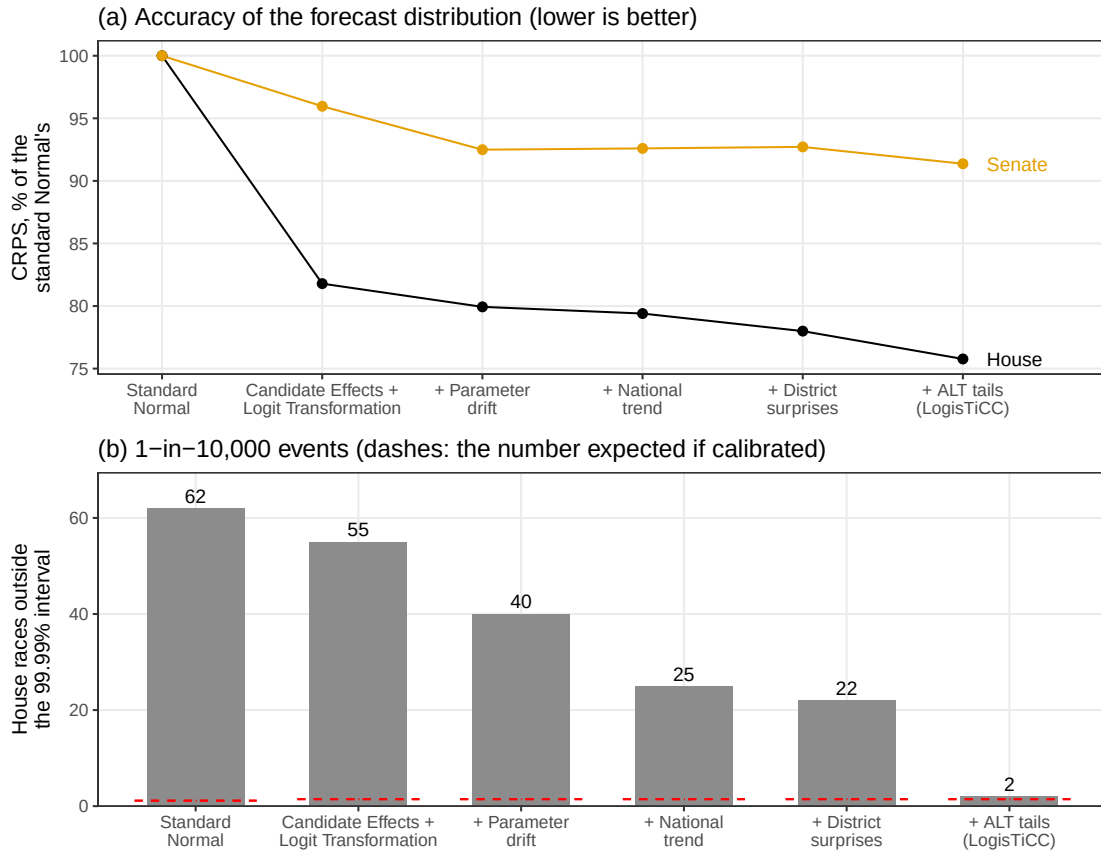


Figure S2: The model built up one component at a time, in the order of Section 2.2.1 of the paper: held-out one-election-ahead forecasts, 1950–2024 (the Senate’s from 1960). (a) CRPS as a percentage of the standard model’s, on the contested races every step forecasts (11,436 House, 1,104 Senate); (b) House races outside each step’s 99.99 percent interval, among all the contested races it forecasts (11,436 for the standard model, 14,478 for the others), against the number a calibrated model expects (dashes).

S4 Computation: GPU acceleration

When expanded fully, our model as described in Section 2 contains 21,841 parameters. Coupling this with the large House and Senate data panel, fitting the model using standard CPU based MCMC samplers would be impractical. In fact, smaller versions of this model that have been fit on CPUs with half of the training data took at least days to train. In response to this, we exploit parallelism in our model’s estimation process and move the computation to an NVIDIA H200 GPU, accelerating posterior estimation from days to 11.0 hours. We now describe the computational process.

Our model is estimated using Hamiltonian Monte Carlo (HMC) using 64-bit floating

point arithmetic for precision and the No-U-Turn Sampler (NUTS). HMC is recursive by nature, but the efficiency bottleneck is not in the recursion, but rather the billions of evaluations of the log-posterior $\log p(\Theta)$ and its gradient $\nabla \log p(\Theta)$ at proposed parameter values. We target this bottleneck by identifying sets of simple, repeated computations during posterior estimation, and packaging them to be fused into large and efficient GPU kernels. The size of an H200 GPU enables these kernels to be run quickly (e.g., gradient evaluations can be spread across thousands of threads in parallel), accelerating each step of the NUTS algorithm.

S4.1 Estimation Tools

We do not write GPU kernels directly. Instead, we leverage PyMC, JAX, and NumPyro to construct our model in a way such that highly efficient GPU kernels are automatically generated during compilation. The tool which we work with most directly is PyMC, which is where we construct a PyTensor computation graph for our model. This graph details the exact model architecture, and is optimized for compilation to GPU kernels. Afterwards, JAX turns the PyTensor graph into a pure function for the log-posterior, enabling automatic-differentiation of each computation feeding the log-posterior and ensuring that all 20 independent chains are evaluated in parallel. Following, XLA lowers that function to GPU machine code and NumPyro’s NUTS sampler begins sampling on the GPU (see Supplementary Appendix [S1.7](#) for sampler settings).

S4.2 Optimizing the Computational Graph

An H200 GPU can be partitioned into a compute section and a memory section. The compute section consists of all of the streaming multiprocessors as well as the computational cores nested within them. This section is responsible for executing all computations. On the other hand, the memory section consists of all forms of memory (e.g., VRAM) which are responsible for storing data. A GPU operates by executing kernels, which are small programs that (generally) read data from memory, bring that data into the compute section, execute the desired computations, and then write the results back to memory. For our purposes, the arithmetic that takes place on the compute section is extremely fast, and the

bottleneck is the read and write operations to and from memory. Therefore, we seek to cut down on the number of kernels we need to launch by fusing together multiple operations into a single kernel. For our model, we achieve this fusion by optimizing our packaging of the model’s PyTensor computation graph, which we detail below.

- **The location as a matrix–vector product.** The fixed-effect component of μ_{it} (3) is a sum of about a dozen coefficient-times-column terms. By expressing this component as a matrix-vector product instead of a sum of terms, we can fuse what would be many small and sparse kernels into a single dense kernel. We apply this same technique to all other sections of our model, whenever possible.
- **Cumulative sums for the random walk.** Instead of expressing a first order random walk as a large chain of dependent additions, we use recursion to express it as a single cumulative sum, enabling the compiler to fuse all of the additions into a single kernel. We use this for the year-varying covariate slopes and for the lagged-vote slope.
- **Indexed writes rather than loops.** Two places splice a value into an existing array, the national-trend column of the year effects, and the imputed lagged-vote values at the missing rows (S8). We do these as single in-graph indexed writes, so they stay inside the compiled graph and avoid breaking the computation graph into separate steps.
- **Elementwise stacks for the scale and tail.** The scale ϕ_{it} (4) and degrees of freedom ν_{it} (5) are each a baseline with a few additive adjustments attached before exponentiation, which again is a stack of operations that can be fused into a single kernel instead of a kernel per operation.

Each of these optimizations reduces the number of kernels we need to launch, thereby reducing the number of memory reads and writes and consequently accelerating posterior estimation.

S4.3 A custom likelihood for uncontested races

As described in Section 2.2.4, an uncontested race contributes not a probability density evaluated at the observed vote, but rather an interval-censored integral. For a Student- t that integral has no closed form that is both numerically stable and easily differentiable. Therefore, we leverage the fact that a Student- t can be expressed as a scale mixture of Gaussians: conditional on the latent scale, the interval-censored probability is a difference of two Normal CDFs, and averaging over that scale yields the desired value. This average is a smooth one-dimensional integral, which we evaluate by 64-point Gauss–Legendre quadrature, a numerical integration technique. The benefits of this design are twofold: this combination of operations is well suited for XLA to collapse it into a single kernel, and because every piece of this design is differentiable, we can compute its exact gradient during sampling.

S5 Alternative Modeling Assumptions

We tried to eliminate any feature of our model not required for accurate out-of-sample validation and accurate uncertainty intervals, to include additional features that would improve performance, and to consider alternative specifications that might be easier to understand.

As we have shown in the main text, the linear-normal model is poorly calibrated for congressional elections. Additionally, we fit a linear-normal Student- t , which failed because it lacked the flexibility and asymmetry in the tails provided by the additive logistic t (ALT). The additive logistic normal failed because it could not properly capture the observed levels of concentration, nor did it accurately capture surprises with appropriate tails. Fitting an IID ALT, that is without contemporaneous correlations, is not well-calibrated because it misses the correlations due to year-to-year swings in the national trend or dependence due to the stability of coefficient estimates; the ablation studies of Supplementary Appendix S2 quantify exactly this degradation.

We also tried other flexible distributions. We tried the Beta distribution, which models the unit interval directly, but produces poorly calibrated results because it, like the IID

normal, does not capture appropriate levels of concentration or tail behavior. We also tried mixture distributions and errors which, while flexible, wound up being highly model dependent, poorly identified, and computationally fragile.

We also attempted to find alternative correlation structures, besides time mixed effects and district random effects on the logit scale, such as regional mixed effects. Besides districts in the south and outside the south, there was little predictable inter-regional variation. Districts in the North, West, and Southwest do not seem to systematically vary, conditional on the covariates. Our covariates include an indicator for districts in the South to capture what appear to be the most important systematic effects. In terms of covariate selection, we made choices for easy comparison to the literature. Our general model structure, like the normal, can easily accommodate other indicators when desired.

S6 Imputation for Uncontested Seats

Missingness due to uncontestedness is an important feature of historical congressional election data. Figure S3 shows the historical rate of uncontestedness in U.S. congressional elections from the model’s own training panel, which ranges from 21 percent of House races in 1950 to 6.3 percent in 2020. Rather than drop these races, which compose a nontrivial share of the data in any given election year, we impute their effective vote shares within the model, as described in Section 2.2.4 of the paper: an uncontested race contributes a censored-interval likelihood term, so the winner is constrained to win, and the missing lagged vote that would otherwise propagate is imputed structurally from the same posterior. We know uncontested vote share data are not missing at random, and the censoring treatment encodes exactly that.

S7 Priors Do Not Drive the Results

We employ weakly informative priors for estimation convenience. Because the panel supplies an average of about 1,800 contested elections per decade, the model does not require regularization to identify its structural parameters, although weakly informative priors reduce computational time for Hamiltonian Monte Carlo convergence. Figure S4

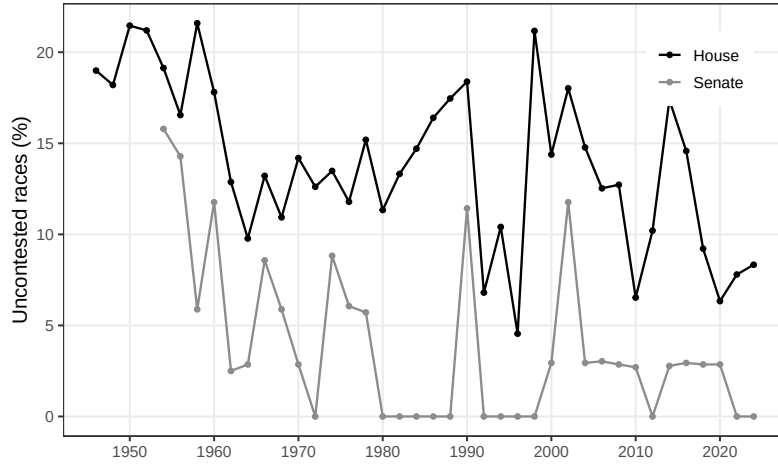


Figure S3: Uncontested elections over time, House and Senate.

overlays the prior and posterior densities for six structural parameters spanning every part of the model: the coefficient on the lagged effective vote, a down-ballot baseline coefficient, the intercepts of the log-scale and log-degrees-of-freedom regressions, the candidate-effect standard deviation, and the polarization coefficient on the scale. In every panel the prior is diffuse relative to the posterior; the data compress the candidate-effect standard deviation to a posterior standard deviation of 0.002 against a prior scale of 0.08, and the down-ballot coefficient to 0.005 against a prior scale of 0.20.

As a direct check that the hyperprior choices do not drive the empirical results, we re-fit the model with the prior scales of the dispersion and hierarchy parameters doubled: the candidate, national-trend, and random-walk scales σ_γ , $\sigma_{\eta,d}$, and σ_{β_k} ; the decade effects ζ_d and α_d ; the degrees-of-freedom intercept α ; and the open-seat, polarization, and single-district/Senate slopes in δ and λ . The scale intercept ζ and the imputation priors keep their scales, and the likelihood and location priors are unchanged. Across the 64 scalar structural parameters the median posterior-mean shift is 0.12 posterior standard deviations, and the location and scale models are essentially unchanged. The only visible movement is inside the degrees-of-freedom block, the component least identified by any finite electoral record: the baseline rises slightly while the covariate loadings deepen (for example, the polarization loading moves from -0.97 to -1.72), offsetting redistributions within the same qualitative structure. The held-out RMSE and CRPS of the doubled-prior fit match the published model’s (the “Hyperprior scales $\times 2$ ” row of Table S6), which is

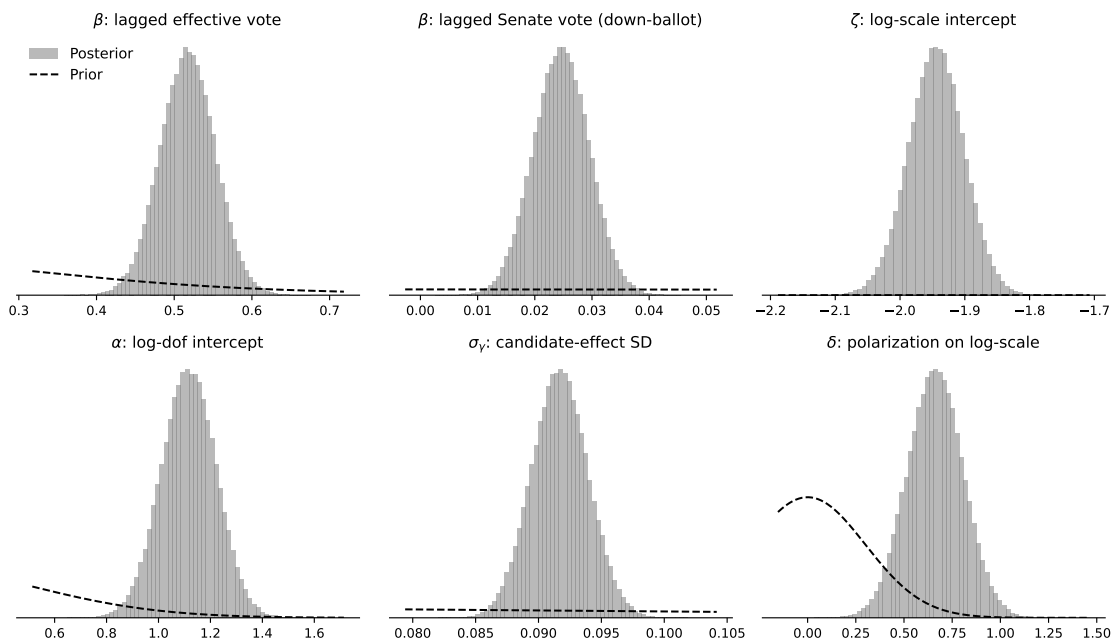


Figure S4: Posterior histograms (grey) against prior densities (dashed) for six structural parameters of the fitted model.

the operative sense in which the priors do not drive the results. Its intervals are somewhat wider: they cover 56.5, 85.7, and 97.6 percent of House outcomes at the 50, 80, and 95 percent levels, against 52.1, 82.2, and 96.1 percent for the published model.

S8 Posterior Estimates

Figures S5–S9 summarize the posterior of the model fit to every election of 1946–2024 (Supplementary Appendix S1.7), with posterior medians and 50 percent (thick) and 95 percent (thin) intervals. Figure S5 gives every constant parameter and the average of each year-varying slope; Figures S6 and S7 show the parameters that change over time, by election and by redistricting decade; Figure S8 shows the concentration that the decade parameters imply; and Figure S9 shows the distribution of the candidate effects.

S9 Computational Details

The standard approach is usually estimated with a linear regression for forecasting (i.e., dropping γ_i) or, for other quantities of interest, via an approximate two-step procedure

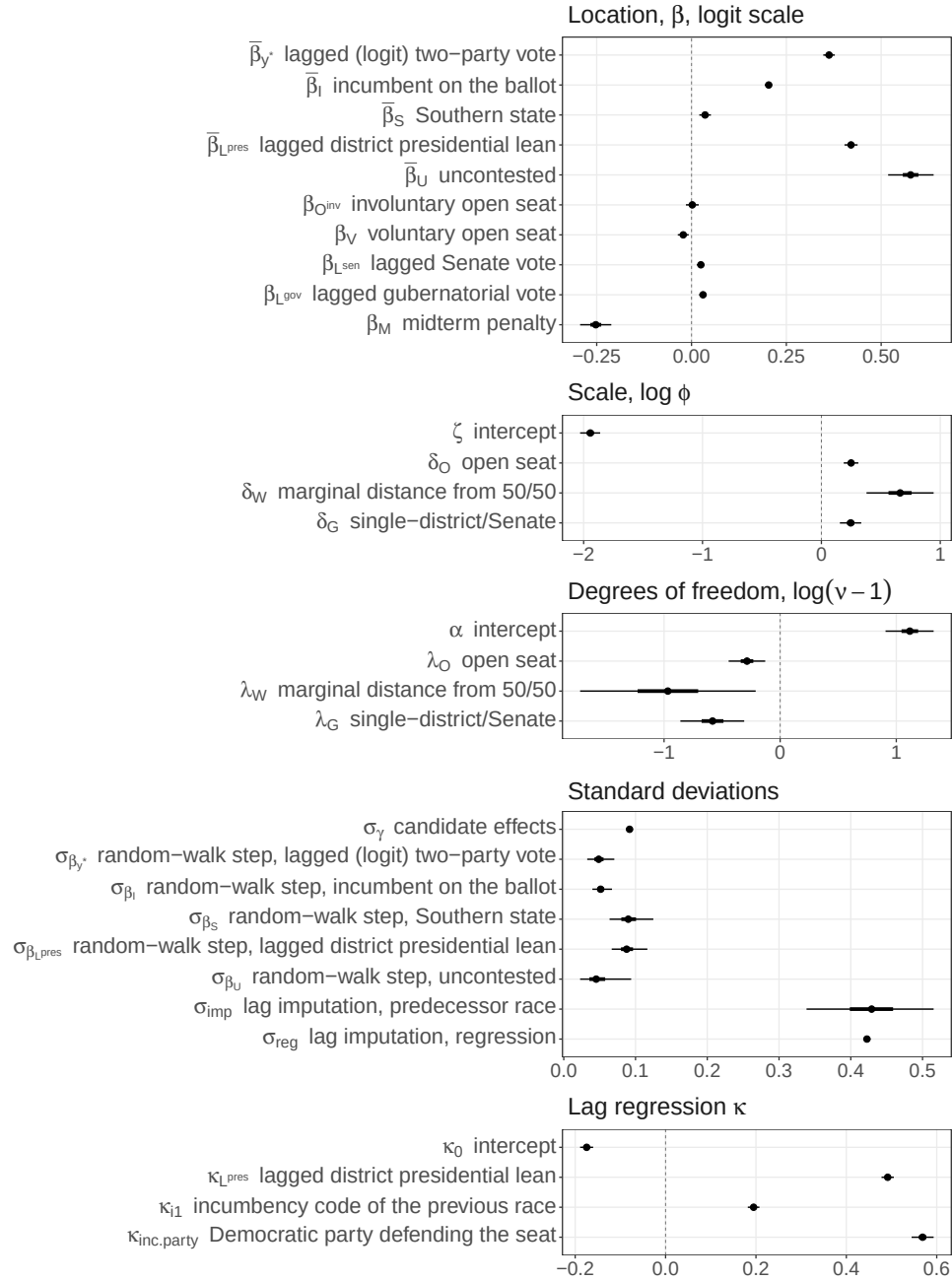


Figure S5: Posterior estimates of the full-data fit, 1946–2024. Location coefficients are on the logit scale, with the five year-varying slopes averaged over the 40 elections and marked with a bar, $\bar{\beta}_k$ (Figure S6 shows them by year). The scale and degrees-of-freedom rows are the intercepts ζ and α and the slopes δ and λ of $\log \phi_{it}$ and $\log(\nu_{it} - 1)$ on the open seat O, the marginal distance from 50/50 W, and the single-district/Senate indicator G (4)–(5). The standard deviations are those of the candidate effects (σ_γ), the random-walk steps (σ_{β_k}), and the two lag imputations (σ_{imp} and σ_{reg}); the last block holds the coefficients κ of the regression (S9) that imputes a lag without a predecessor race. The party premiums β_{A_d} and β_{E_d} and the decade effects appear in Figure S7.

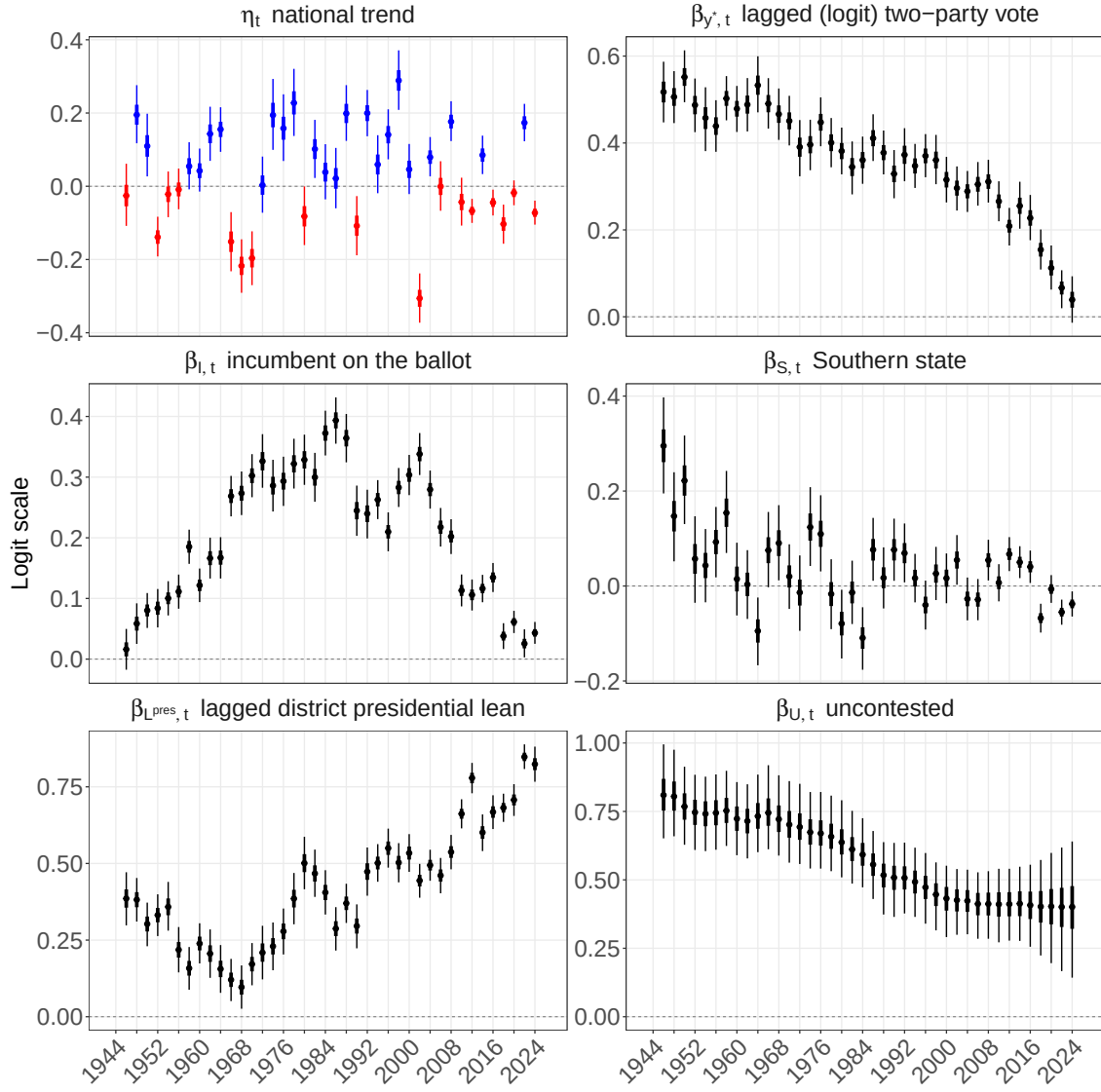


Figure S6: The national trend η_t and the five year-varying slopes β_{tk} in each election, full-data fit, 1946–2024. Each walk starts at its grand mean in 1946. The national trend is blue where its posterior median favors Democrats and red where it favors Republicans.

designed to avoid computational challenges that were difficult in the 1990s (see Gelman and King, 1994).

Because of improvements in computation and Bayesian modeling, we estimate the LogisTiCC via a fully Bayesian specification, sampling the posterior with the No-U-Turn Hamiltonian Monte Carlo sampler on GPU hardware (see Supplementary Appendix S4). The production fit runs 20 chains of 8,000 draws each after 4,000 warmup iterations at a maximum tree depth of 13, and completes on a single NVIDIA H200 in roughly

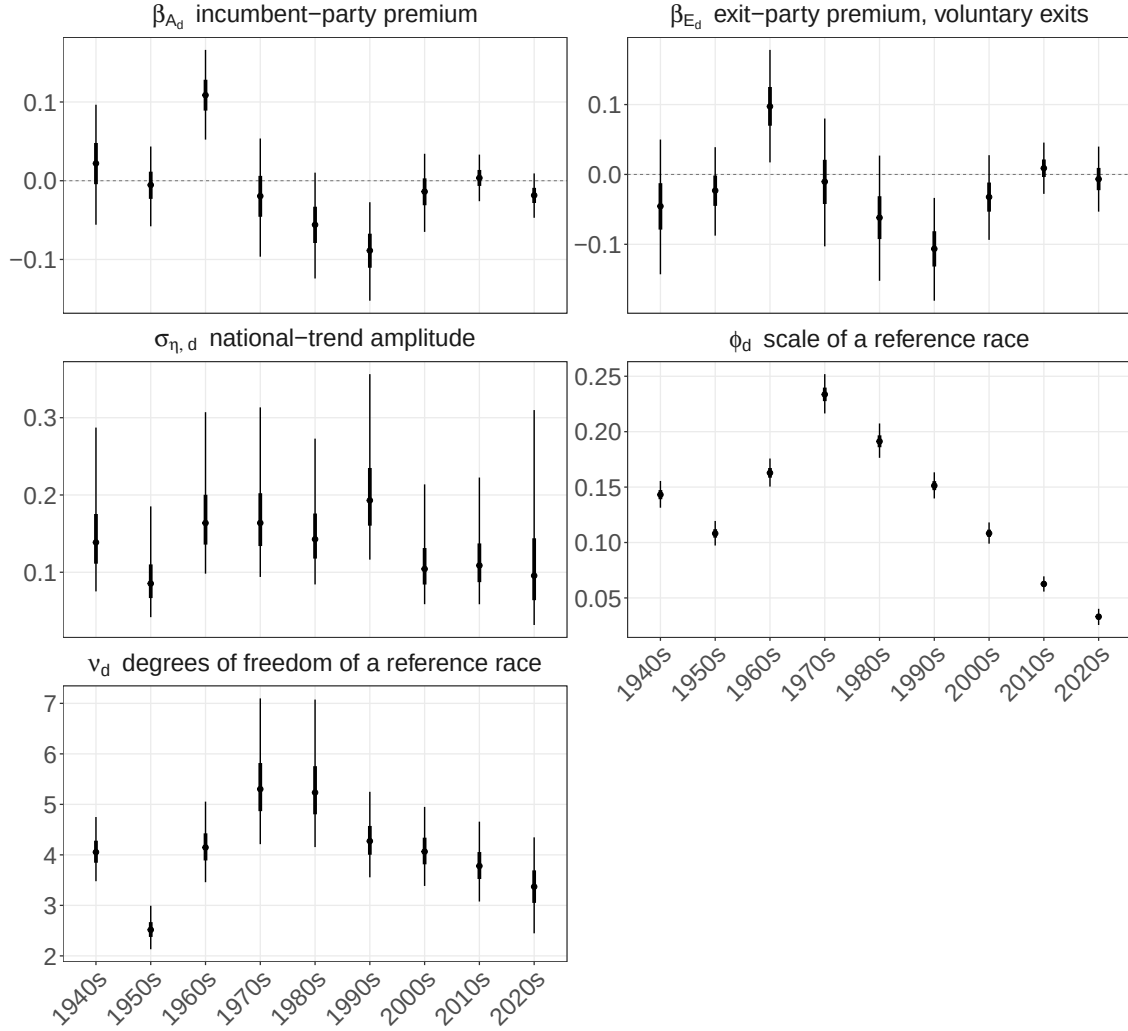


Figure S7: The parameters that vary by redistricting decade, full-data fit: the incumbent- and exit-party premiums β_{A_d} and β_{E_d} (E_d marks seats left voluntarily; the involuntary open seat O^{inv} is in Figure S5), the national trend’s standard deviation $\sigma_{\eta,d}$, and the scale $\phi_d = \exp(\zeta + \zeta_d)$ and degrees of freedom $\nu_d = 1 + \exp(\alpha + \alpha_d)$ of a reference race, an incumbent running in a 50/50 district of a state with several districts.

eleven hours; the full out-of-sample evaluation, which re-fits the model once for every held-out election year, requires about 73 GPU hours. All chains converge, with $\hat{R} \leq 1.01$ across parameters, no divergent transitions beyond a handful in hundreds of thousands of draws, and well-mixed chains. We are able to simulate quantities of interest directly from the full joint posterior distribution of the predicted values and parameters, which means researchers can easily calculate any relevant quantity of interest, along with accurate and calibrated uncertainty estimates.

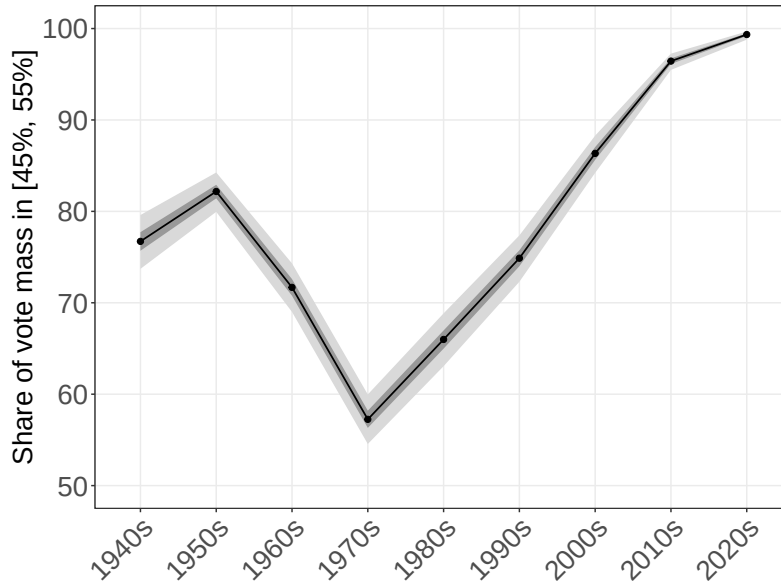


Figure S8: Concentration by redistricting decade, full-data fit: the probability that the reference race of Figure S7, with an expected vote of 50 percent, ends between 45 and 55 percent, $2F_{t_{v_d}}(\text{logit}(0.55)/\phi_d) - 1$. Posterior median with 50 and 95 percent intervals.

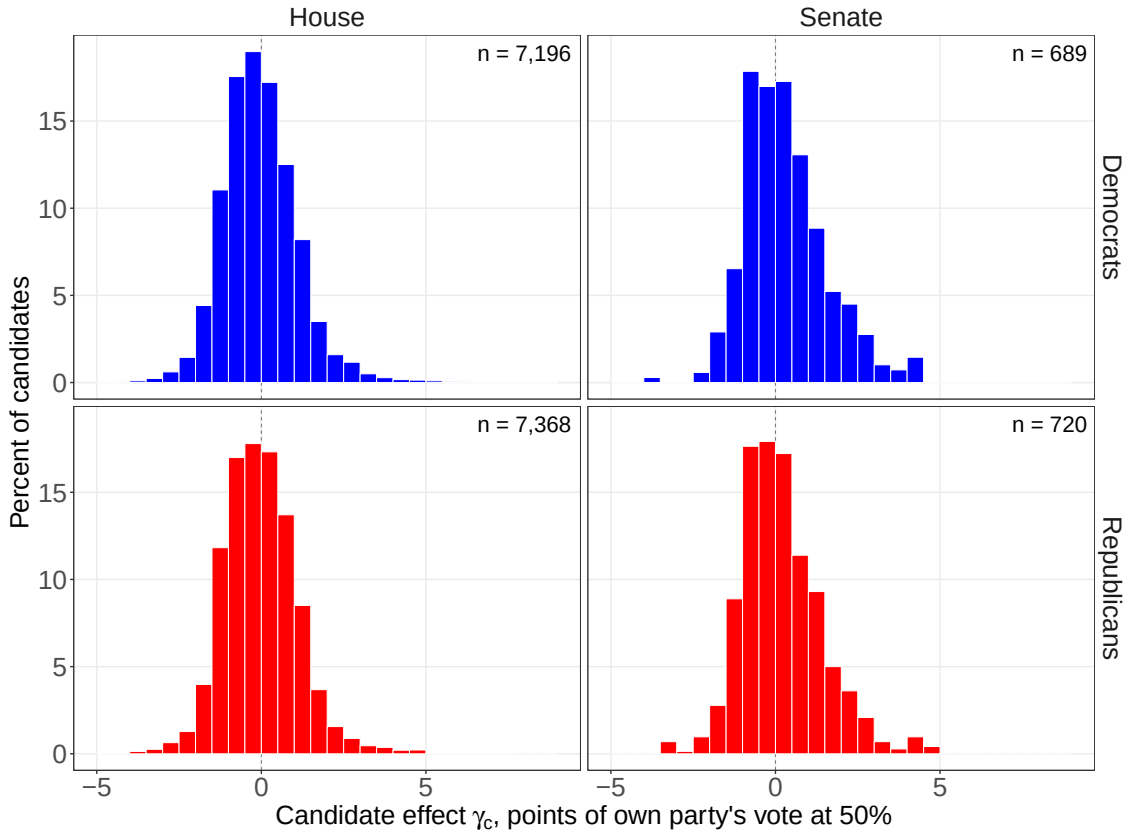


Figure S9: Posterior means of the 15,973 candidate effects γ_c , full-data fit, in percentage points of the candidate's own party's vote at 50 percent, by party and the chamber of the candidate's last race. Posterior means shrink toward zero for candidates with few races.

S10 Predictive Accuracy by Standard Metrics

We evaluate the LogisTiCC and the Normal baseline on the same strictly out-of-sample forecasts used in Section 3 of the paper: for the LogisTiCC, the leave-future-out hold-out forecasts (each election predicted from a model fit only on prior elections); for the Normal, the every-election one-step-ahead refit described in Section 3.3. Table S7 reports root mean squared error and mean absolute error of the predicted two-party vote share, the continuous ranked probability score (CRPS; Gneiting and Raftery 2007), and empirical coverage of the central 50, 80, and 95 percent predictive intervals, computed on the common set of contested races.

The LogisTiCC improves on the Normal baseline on every metric in both chambers. The point predictions are more accurate (House RMSE 0.069 against 0.086), the full predictive distributions are sharper where it counts (CRPS 0.037 against 0.048), and the intervals attain close to nominal coverage at every level, where the Normal is badly overconfident (House 80 percent intervals cover 69 percent of outcomes; 95 percent intervals cover 87 percent). Because the CRPS integrates over the entire predictive distribution, it rewards both calibration and sharpness; the LogisTiCC’s advantage on CRPS shows its intervals are not wide indiscriminately but are wide precisely where the data require width.¹

	House		Senate	
	LogisTiCC	Normal	LogisTiCC	Normal
RMSE (vote share)	0.069	0.086	0.089	0.096
MAE (vote share)	0.052	0.068	0.067	0.075
CRPS	0.037	0.048	0.048	0.053
50% coverage	51.9	40.0	47.5	47.6
80% coverage	81.6	68.5	77.6	77.5
95% coverage	95.9	87.3	93.4	92.4
Races (contested, common set)	11,436 (1950-2024)		1,104 (1960-2024)	

Table S7: Out-of-sample predictive accuracy, LogisTiCC against the Normal baseline, on the common set of contested races. Vote shares on the [0,1] scale; coverage in percent.

¹The LogisTiCC’s holdout reports store the predictive distribution as a ladder of 20 quantiles; we compute the CRPS for both models by the standard pinball-loss quadrature over the same quantile grid, so the comparison is exact like-for-like.

S11 Comparison of Nominal Confidence Interval Lengths

To quantify the magnitude of uncertainty differences between the Normal and LogisTiCC models for district- and legislature-level statistics, we compute the ratios of the credible interval (CI) widths from these two models. For each out-of-sample race we compute the width of the 95% predictive interval under both models and take the ratio of the LogisTiCC width to the Normal width. Figure S10 reports the distribution of these ratios.

At the district level the two models produce intervals of comparable width: the mean ratio is 1.07 in the House (standard deviation 0.21) and 0.98 in the Senate (standard deviation 0.22). The LogisTiCC does not fix the Normal model's coverage failures by inflating every interval; it reallocates width across races through the modeled scale and degrees of freedom, widening where races are volatile and narrowing where they are not, which is why it achieves nominal coverage at similar average width (Table S7). At the legislature level, by contrast, the positive cross-district correlations induced by the national trend term η_t widen the LogisTiCC's intervals for aggregate quantities such as the median seat substantially relative to the independence-assuming Normal model.

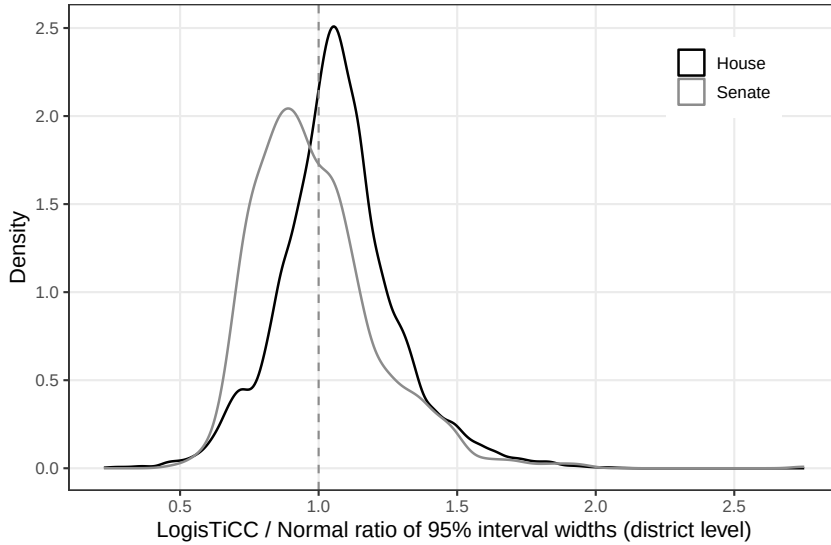


Figure S10: LogisTiCC-to-Normal ratios of 95% predictive interval widths at the district level, out of sample.

S12 Additional Calibration and Coverage Results

The out-of-sample coverage of the LogisTiCC’s 95 percent intervals, under the leave-future-out design of Section 3.2 of the paper, closely tracks their in-sample coverage in the fit to all elections (Figure S11). Figure S12 shows that the LogisTiCC model calibrates well for each of the redistricting decades.

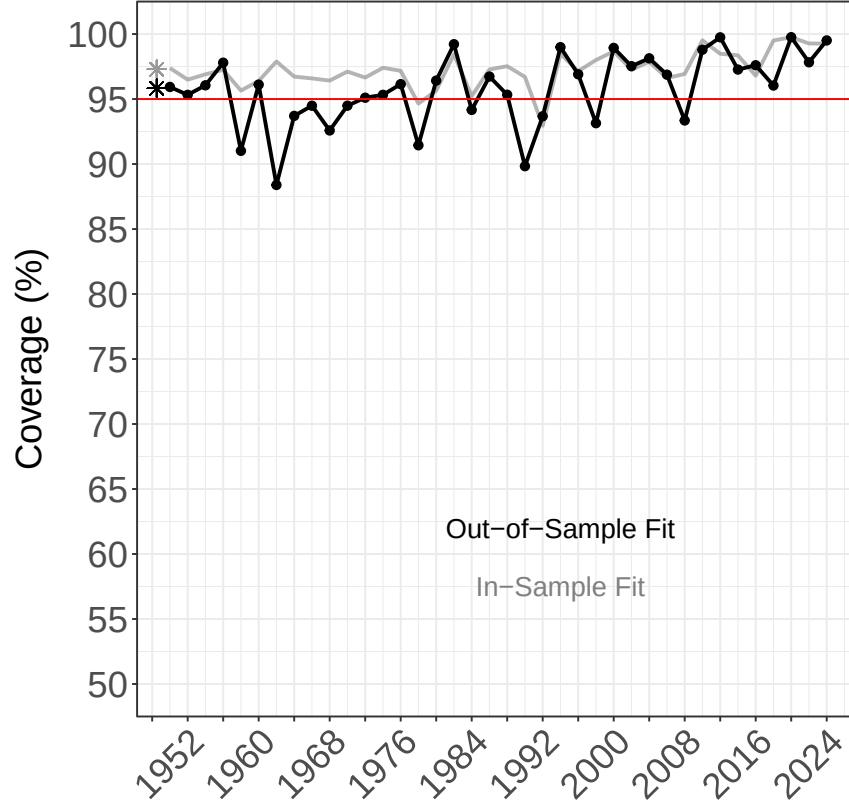


Figure S11: **Out-of-sample against in-sample coverage.** Empirical coverage of the nominal 95% forecast interval by year under the leave-future-out design described in Section 3.2 (train on prior elections through $t - 2$, forecast year t , 1950–2024). The lighter series is the in-sample reference from the full-data fit; the out-of-sample forecast tracks it closely. Stars to the left of the first year mark each series’ average coverage across years, weighted by the number of contested races (black: out-of-sample; gray: in-sample). Both series are reproducible from committed forecasts. The in-sample reference is computed from the full-data fit at 100,000 draws.

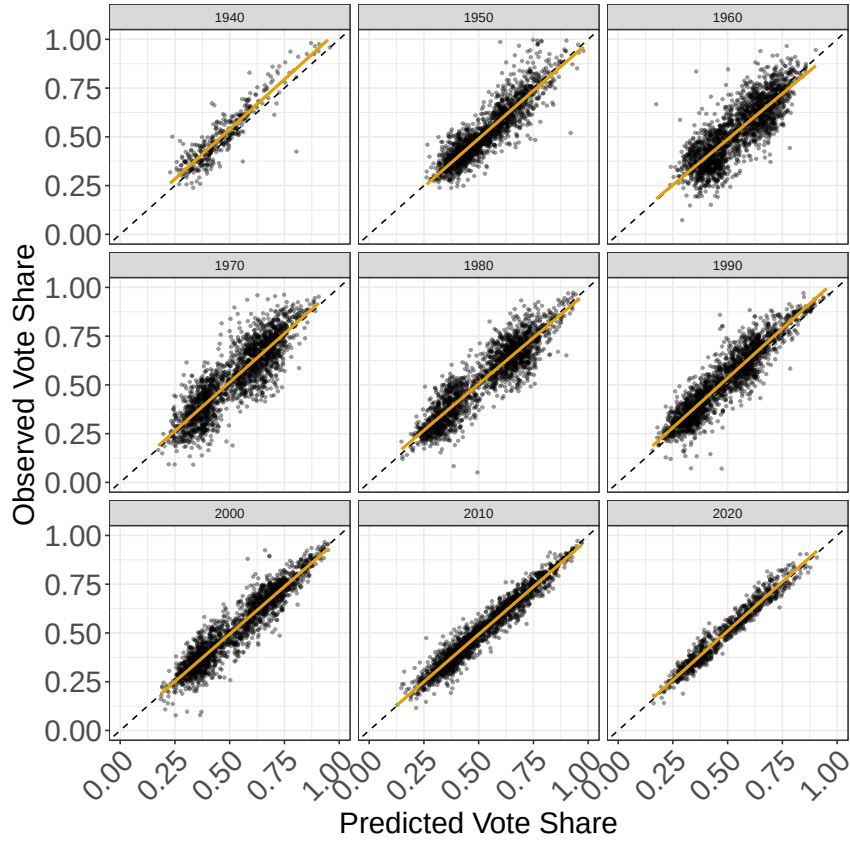


Figure S12: **Calibration by redistricting decade.** Each point is a one-election-ahead out-of-sample forecast: for every election year t from 1950 to 2024, the model is fit to all elections through $t - 2$ and forecasts year t . Panels group these forecasts by redistricting decade and plot the predicted against the observed Democratic two-party vote share. The dashed line is exact calibration (45°) and the solid line a within-decade fit.

S13 The History of Generative Modeling

To calculate generatively accurate descriptive summaries, the statistical model generating these summaries should (a) pass extensive, rigorous out-of-sample tests that validate its generative abilities and (b) reflect available prior information from the literature. In our efforts to meet these conditions, we benefit from developments in three major fields of statistics, each of which has engaged with these same conditions. We now situate the ideas described in this paper in the history of statistical analyses by briefly describing these three research traditions.

First, direct attempts to build generative models in the social sciences have a long history, from path analysis originating in 1920s sociology, to linear structural equation

modeling in econometrics and psychometrics in the 1960s and 70s, and, more recently, to hierarchical Bayesian models in statistics. At one point, econometricians had built many structural equation models of the economy, sometimes with hundreds of equations and each finely tuned to their past theoretical knowledge. However, rigorous out-of-sample forecasting were surprisingly embarrassed by a comparison with “atheoretical” univariate ARIMA models, leading many to reassess the value of their prior information. These attempts failed because researchers lacked the requisite computational resources to build models that reflected prior knowledge and sufficient data to make extensive validation possible. Now, model checking has become a more common part of Bayesian best practices (e.g., Gelman, Meng, and Stern, 1996).

Second, when estimating accurate generative models was not feasible, or required too many unjustified assumptions, social scientists turned to other research frameworks, often changing their quantities of interest in the process. Most notably, the literature on causal inference, especially since the 1980s, has made tremendous progress by developing ways of estimating causal effects without modeling assumptions. Although numerous articles had previously attempted to make causal inferences, Leamer (1983) and others pointed out that high levels of (what came to be known as) model dependence meant that most of these inferences were not right, and maybe not even wrong, but instead mostly reflected researchers’ priors. The “credibility crisis” that resulted from this skepticism and from rigorous tests of observational estimates compared with out-of-sample randomized experiments (Lalonde, 1986), lit a fire under the methodological community, resulting in remarkable progress that continues until today (Imbens, 2022). The theories and descriptive stories that emerge from generative models, including ours, often include many causal effects, and so the ability of these methods to proceed without modeling assumptions has been valuable for everyone. At the same time, even if we had exact knowledge of all causal effects ever estimated and a vast number of others, we would not come close to the range of descriptive knowledge social scientists seek and which can be gained by generatively accurate descriptive summaries. Descriptive quantities such as partisan bias, responsiveness, forecasts, farcasts, and many others are not causal effects but of course

remain of interest to political scientists and policymakers.

Finally, machine learning methods of classification and prediction have made continual progress by their single-minded focus on out-of-sample validation. By taking their task as engineering better algorithms and downplaying constraints suggested by prior theoretical “knowledge,” they make themselves continually vulnerable to being proven wrong. Although one can often do as well with simpler models that explicitly code more prior knowledge, this literature’s focus on validation helps them avoid being fooled by elegant theories that do not have empirical support.

As has been true throughout the history of quantitative social science methodology, political scientists have a comparative advantage when they employ their knowledge of the political world, but do best when subjecting their statistical claims to the possibility of being proven wrong.

S14 Uses of AI

This appendix reports on our use of large language models (“AI”) for this project, by period and task, with the tools and versions at the end. All dates are 2026.

- **1/17–4/9, OpenAI Codex.** Debugged the forecast function for error propagation and conditioning sets (1/17; `manual_forecast_re()`, whose Python port is now the forecast engine of the replication package). Built the lagged statewide votes: Senate-with-House preparation and statewide lag averages (2/5, 2/19); a search for district-level presidential, Senate, and governor returns in the CQ Almanac and ICPSR holdings, with county-to-district crosswalks and parsers for California and Wisconsin (2/23); county-level governor and Senate results for 1960–2020 compiled to districts by `build_cd_level_results.R` (2/25), stage 2.1 of the data-creation pipeline. Model code: a Stan version with an R front end (2/28–3/1); diagnostics on the FASRC cluster (3/8); joint House–Senate runs and plots (3/11–14); an early version of the chamber-interaction model (3/18); audits of the 2020 calibration and of the lags (3/20–23). Text: a results document with drafted figure captions (3/16–17); a paragraph on the imputation steps and data sources,

with data citations checked against their web sources (3/25–26).

- **2/9, GitHub Copilot Chat (Claude Haiku 4.5).** A combined House–Senate script; a search for district-level governor and Senate returns back to 1956.
- **3/14–4/9, Claude Opus 4.6.** Ported the brms and Stan model to the PyMC and JAX GPU code, checking each prior against the Stan code; translated the R prediction pipeline to Python (the model code records that it was translated from those R functions); diagnosed calibration problems and slow sampling, with notes in the debug directory of the GPU code. Eight commits naming the model (3/27–4/8) restructured the repository and added the README, analysis plots, LaTeX figures, and the one-in-10,000 Senate plot.
- **3/24, Claude Sonnet 4.6.** Edited a PolMeth proposal in LaTeX.
- **4/24–6/30, Claude Opus 4.7, then Fable 5, in the GPU code.** Audited the GPU forecast against the R version and fixed over-coverage (4/24); implemented the imputation model for redistricted seats (5/7); calibration work and the Senate member-ID mapping (5/19–20); specified and tested model variants M54–M59, rewrote the forecast algorithm, and corrected the 1966 and 2010 midterm coding (5/23–25), after which M59 became the production model; forecast-time dispersion terms (6/10).
- **6/25–8/5, OpenAI Codex.** Drafted two sections and one appendix of an earlier version of this Supplementary Appendix (6/25–28), with Codex reviews and fixes.
- **7/3–10, Claude Fable 5.** Grammatical edits and citation fixes.
- **7/3–8/17.** About 40 merged pull requests to the main paper’s source: copyedits of the introduction and model sections; rewrites of the calibration text; the appendices on statistical details and priors, the priors tables, and figure fixes.
- **8/7–10/4, Claude.** Drafting, number updates, audits, the Supplementary Appendix, and the bibliography. Sixteen commits naming the model (8/13–10/2) built the replication package.

- **9/30, ChatGPT.** Review of the paper.
- **10/7, Claude.** Compiled this record.

Tools and versions.

- **Claude (Anthropic), through Claude Code.** Opus 4.6 (3/14–4/9); Sonnet 4.6 (3/24); Opus 4.7 (4/24–6/30); Fable 5 (6/10–9/9); Opus 4.8 (6/16–7/1, on a companion project only); Opus 5 (7/29–9/10); Fable 5.1 (9/10–23); Opus 5.5 (9/23–10/7). Claude Code versions 2.1.170–2.1.288 from May on; earlier versions were not recorded.
- **OpenAI Codex.** 1/17–4/9: the VS Code extension (0.81–0.98), then the Codex desktop app (0.95–0.118), with the models GPT-5.2-codex, GPT-5.3-codex and its -spark variant, GPT-5.4, and GPT-5.5. 6/25–8/5: version not recorded.
- **GitHub Copilot.** Copilot Chat in VS Code, 2/9 (Claude Haiku 4.5 and automatic model choice); the Copilot CLI, 7/8 (GPT-5.3-codex).
- **ChatGPT (OpenAI).** GPT-6 Astra (high), 9/30.

We reviewed the paper with ChatGPT GPT-6 Astra (high) and Claude Fable 5.1, and collected the information for this list mostly with Claude Fable 5.1.

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